

Rebaz Kamal

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Education

University of Massachusetts Boston

Sept. 2022 - Present

B.S. in Computer Engineering, Minor in Computer Science

- GPA: 3.97/4.0
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Applied Machine Learning, Computer Security, Embedded Systems, Operating Systems, Digital Systems, Statistics & Applications

Experience

University of Massachusetts Boston

Boston, MA May 2025 - Present

Research Assistant, Artificial Intelligence & Machine Learning Lab

- Conduct research in machine learning with a focus on ensemble methods and semi-supervised learning
- Develop and evaluate deep learning models for image classification and segmentation using PyTorch and TensorFlow
- Analyze experimental results and generate visualizations to support research publications and conference presentations

Airion Technologies

Boston, MA Jan. 2025 - Present

Co-Founder & Lead AI/ML Engineer

- Co-founded a startup developing intelligent flight controllers with embedded machine learning for real-time system monitoring and diagnostics
- Designed and trained ML models to analyze flight sensor data and generate predictive health scores for detecting system degradation and potential failures
- Leading development of lightweight AI models optimized for deployment on embedded flight control hardware

University of Massachusetts Boston

Boston, MA Jan. 2024 - Jan. 2026

Teacher Assistant, CS240 (Programming in C)

- Supported 250+ students in learning C programming concepts including memory management, pointers, and data structures
- Assisted with debugging, code optimization, and developing structured problem-solving strategies
- Provided technical feedback on lab assignments to improve programming practices and code readability

University of Massachusetts Boston

Boston, MA Jan. 2026 - Present

Teacher Assistant, ENGIN246 (Computer Organization and Assembly)

- Supported 40+ students in understanding computer organization concepts including CPU architecture, instruction set design, and computer arithmetic
- Assisted with assembly language programming assignments and debugging low-level code
- Provided guidance on datapath design, pipelining, and memory hierarchy concepts through lab support and technical feedback

University of Massachusetts Boston

Boston, MA Jan. 2025 - May 2025

Teacher Assistant, ENGIN241 (Digital Systems with Lab)

- Assisted 30+ students in designing and simulating digital circuits using combinational and sequential logic
- Supported FPGA-based development using VHDL, including simulation, synthesis, and hardware debugging
- Provided lab guidance and feedback to improve understanding of digital system design principles

Revere Public Schools

Revere, MA Jan. 2023 - June 2023

Technology Intern

- Diagnosed and resolved hardware and software issues for 500+ staff and student devices

- Configured and deployed technology equipment to maintain reliable and secure system performance
- Documented troubleshooting procedures and collaborated with IT staff to support district-wide infrastructure

Research & Publications

CLESP: Collaborative Learning with Ensemble Soft Pseudo-Labels

Manuscript in Progress

- Co-authored a novel ensemble learning method for test-time adaptation that enables classifiers to collaboratively improve using soft pseudo-labels
- Proposed a dynamic pseudo-labeling strategy allowing models to learn from the highest-confidence ensemble member without labeled data
- Demonstrated improved generalization under distribution shift on CIFAR-10, CIFAR-100, and ImageNet-C datasets

Reliability-Aware Knowledge Distillation for Language Models *In Preparation (Target: EMNLP 2026)*

- Investigating the impact of symbolic vs. soft supervision on student model reliability and confidence calibration
- Evaluating performance using accuracy and ROC-AUC to measure confidence-correctness alignment
- Exploring intermediate (local) loss functions to improve robustness and failure detection

Projects

AI-Integrated Flight Controller for Predictive System Health Monitoring

- Developed an embedded AI system for real-time fault detection and predictive health monitoring of drone components including battery, ESC, motor, and propeller
- Designed models using sensor data (voltage, current, RPM, PWM, temperature, IMU) to detect failures and generate a system-wide health score (0–100)
- Enabled early failure detection and preventive actions, reducing operational risk, maintenance costs, and in-flight system failures

Lateral Rapid Diagnostic Machine (L.R.D.M.) – AI-Enabled Medical Diagnostic Device

- Developed a portable diagnostic device for real-time disease detection using lateral flow assays and automated image-based analysis
- Integrated machine learning with hardware (PCB design, sensors, imaging) to classify test results and enable low-cost point-of-care diagnostics
- Collaborated on a multidisciplinary engineering team to design a scalable and accessible medical testing system

Skin Cancer Classification using Convolutional Neural Networks

- Developed a CNN-based model for skin lesion classification using transfer learning, achieving **86% test accuracy** on the HAM10000 dataset
- Implemented data preprocessing, augmentation, and class balancing to improve classification performance
- Evaluated model performance using classification metrics and threshold tuning to improve malignant detection

Leadership & Activities

UMass Boston Robotics and Engineering Club

Boston, MA Sept. 2025 – Present

Vice President

- Lead hands-on workshops teaching students soldering, CAD, and embedded programming using Arduino and Raspberry Pi platforms

- Work on club engineering projects including drone systems and microcontroller-based robotics applications
- Preparing and leading the team to compete in the MassRobotics Form & Function Challenge against universities including MIT and the University of Waterloo

Paul English Applied Artificial Intelligence Institute (PEAII)
AI Institute Assistant

Boston, MA Jan. 2026 – Present

- Help organize AI workshops, hackathons, and technical events hosted by the Paul English Applied Artificial Intelligence Institute
- Support the planning and facilitation of AI Summer Camps introducing students to machine learning, programming, and applied AI tools
- Assist with outreach initiatives that connect students with AI research, industry professionals, and career opportunities

Skills

Programming Languages: Python, R, SQL, C/C++, Java, MATLAB

Machine Learning Frameworks: TensorFlow, PyTorch, Scikit-learn

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Tableau, Power BI

Development Tools: Jupyter Notebook, VS Code, PyCharm, IntelliJ, Git, GitHub

Awards/Honors

The National Grid Clean Energy Academy Scholarship | Issued by **National Grid**

Alpha Lambda Delta (ALD) National Honors Society | Issued by **Alpha Lambda Delta**

Dean's List | Issued by **UMass Boston** (each semester)

Certifications

Data Engineering, Big Data, and Machine Learning on GCP Specialization | **Google Cloud**

Deep Learning Specialization | **DeepLearning.AI** (Stanford University)

Machine Learning Specialization | **DeepLearning.AI** (Stanford University)